

CSA15 Cloud Computing and Big Data Analytics

CO4 AT1 - Big Data Analytics Class Test Student Template

Course Code	CSA15	Slot	B
Course Title	Cloud Computing and Big Data Analytics	Assessment	CO4 AT1
Assessment Title	Big Data Analytics Class Test	Bloom Level	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Submission Mode	Individual digital answer template through GitHub and Google Classroom	Total Marks	30

Student Details

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Class / Section		Slot	B
Capstone Title	SecureVote: A Green Cloud-Based Campus Election Management System Using Audit Logs and Result Integrity	Submission Date	19-8-2026
GitHub Repository Link		Google Classroom Status	SUBMITTED

Assessment Overview

CO4 Focus

Analyze big data characteristics, processing needs, security challenges, business drivers and domain applications. This template includes explanation notes to guide the thinking process. The final answer must be written in the student's own words.

How to use this template:

- Read each question and the explanation notes before answering.
- Use your capstone product as an example wherever relevant.
- Write concise technical answers; avoid copying the explanation notes directly.
- Use diagrams or small tables wherever they improve clarity.
- Upload the completed DOCX/PDF and supporting evidence to GitHub; share the repository link through Google Classroom.

Assessment Rubric

Criteria	Marks	Evaluation Focus	Student Check
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Concept Accuracy	9	Correct technical meaning, terminology and examples.	Have I used correct big data terms?
Coverage	6	All required parts of the question are answered.	Have I answered every part?
Examples	4.5	Relevant cloud/capstone/domain examples are included.	Have I used meaningful examples?
Analysis	6	Implications, security, scalability and decision value are explained.	Have I explained why it matters?
Clarity	4.5	Answer is organized, concise and readable.	Is my answer easy to evaluate?

Code of Conduct and Student Assurance

I __VISWANATH SAI SANDEEP (192324029)__(Name / Register Number) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____VISWANATH SAI SANDEEP_____

Question and Answer Sections

Answer all questions. Each question carries 3 marks. Use the explanation notes for direction, but write the final answer independently.

Question 1 (3 Marks)

Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Explanation Notes

1	Start with a precise definition: Big Data refers to datasets whose size, speed, variety or complexity exceeds the capability of conventional tools.
2	Mention why spreadsheets or single-server databases struggle: storage limits, slow queries, concurrency issues, schema rigidity and limited scalability.
3	Use one cloud/capstone-style example such as app logs, IoT feeds, transactions, images or clickstream events.
4	End with why distributed storage and parallel processing become necessary.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Definition:

Big Data refers to datasets whose size, variety or complexity exceed the capability of conventional tools for processing. This was famously defined by the 3 Vs (Volume, Velocity, Variety) introduced by Mark Allen Weiss in 2006, which were later refined into the 4 Vs (Volume, Velocity, Variety, Veracity) by Facebook in 2013.

Why traditional approaches fail? Traditional data processing tools such as relational db, spreadsheets & single server systems struggle because they are designed for small, static datasets.

Challenge	Traditional Approach	Big Data Reality
Storage Limits	Single disks / small RAM	Exabytes of data that cannot fit in a database
Slow Queries	Indexes, batch processing	Unstructured data, massive query volumes
Concurrency	Single-threaded / sequential	Hundreds of concurrent users accessing data simultaneously
Schema Rigidity	Pre-defined tables, fixed columns	Unstructured data (images, logs, events) with no fixed schema
Scalability	Limited to a single machine	Exponential growth in data volume, requiring horizontal scaling

Example: A social media platform Instagram

A social media app processes petabytes per day of data.

1. 10 million photos
2. 50 million interactions
3. 500 million users
4. 10 million users

Traditional tools fail here because

1. Storage needed to store all images & metadata far exceeds any single server's RAM or Disk
2. Concurrency – millions of simultaneous logins and feed requests cannot be handled sequentially.

3. Schema flexibility – new content format (videos, stories) constantly evolves requiring no rigid schema
4. Velocity – new posts likes & interactions arrive continuously requiring real time processing.

Because of these constraints, distributed storage (e.g., HDFS, cloud object storage) and parallel processing (e.g., Spark, Flink) become necessary. They allow the system to handle massive data volumes, process data in parallel across many nodes, scale horizontally as data grows, and handle unstructured data without redefining a rigid schema — a necessity that traditional approaches simply cannot satisfy.

Question 2 (3 Marks)

Explain the 5Vs of Big Data with one practical example for any two Vs.

Explanation Notes

1	List the 5Vs clearly: Volume, Velocity, Variety, Veracity and Value.
2	For at least two Vs, connect the definition to a concrete scenario instead of writing only textbook meaning.
3	Example: In a smart traffic system, velocity refers to continuous vehicle movement updates; value refers to congestion prediction.
4	Show that each V affects system design, storage, processing or analytics decisions.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

The 5Vs of Big Data

Overview

The 5 Vs of Big Data are the key dimensions that distinguish big data from traditional data:

V	Value
The value of the data itself	The insights and insights gained from the data
V	Volume
The size of the data set	The sheer scale of data (terabytes, petabytes, exabytes)
V	Velocity
The speed at which data is generated	Speed of data generation (real-time, near real-time)
V	Variety
The types of data (structured, unstructured, semi-structured)	Diverse data sources
V	Veracity
The quality and reliability of the data	Uncertainty in data accuracy
V	Variation
The diversity of data (user behavior, location, time, etc.)	Dynamic changes and diversity

Practical Example: A Social Media Platform

Volume

A social media platform (e.g., Instagram) may collect exabytes of data daily — millions of posts, photos, comments, likes, and location tags. Storing this all requires petabytes of storage.

Velocity

Posts and media are generated continuously (hundreds of thousands per second). Traditional batch processing cannot keep up.

Variety

The data is not uniform — it includes text, images, videos, geo-location, audio, and metadata from users, devices, and third-party services.

Veracity

Content may be fake, misleading, or fabricated (e.g., deepfakes, bot-generated posts, coordinated inauthentic behavior).

Variation

The same user can post different content at different times, from different locations, on different devices.

Why Traditional Approaches Fail Here

Traditional Approach

Single relational database

Single server

Batch processing (hourly/every 6 hours)

Fixed schema

No replication

No fault tolerance

Big Data Challenge

Too little data to justify cost of DB

Not designed for 100+ concurrent users

Real-time decisions needed (e.g., notifications, ads)

Can't handle diverse data types (images, video, text)

Data is lost if one node fails

One server failure = platform outage

Real-World Consequence

Without proper big data infrastructure, the platform would suffer from:

- Data loss during server failures
- Slow response times during peak traffic
- Inability to scale during viral content spikes
- Inability to process images, videos, and unstructured content
- No real-time insights (e.g., trending topics, viral content detection)

This is why distributed storage (HDFS, S3) and parallel processing frameworks (Hadoop, Spark, Flink) are essential.

Question 3 (3 Marks)

Differentiate structured, semi-structured, and unstructured data with examples.

Explanation Notes

1	Structured data has a fixed schema, such as customer table, transaction table or attendance record.
2	Semi-structured data has tags/keys but flexible structure, such as JSON, XML, API responses or log files.
3	Unstructured data has no fixed tabular form, such as images, videos, audio, PDFs, emails or social media text.
4	Use examples from one cloud-backed application to make the comparison coherent.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Structured vs. Semi-Structured vs. Unstructured Data

Quick Comparison

Feature	Structured Data	Semi-Structured Data	Unstructured Data
Format	Pre-defined schema	Partially formatted	No defined format
Examples	SQL tables, spreadsheets	XML, JSON, CSV	Emails, PDFs, social media posts
Parsing	Easy, direct access	Requires parsing	Manual or heuristic parsing
Database type	Relational (SQL)	Document databases, APIs	No database needed
Scalability	Limited	Moderate	High

1. Structured Data

Definition: Data stored in predefined fields with a fixed schema (fixed columns and types).

Examples:

- Customer DB: CustomerID, Name, Age, Address, Email
- Sales Record: OrderID, ProductID, Quantity, Price, Date, Status
- Employee Record: EmployeeID, Department, Salary, HireDate, ManagerID

Why it works: The database engine knows exactly which fields to read and where to look for values. Queries are fast and efficient.

2. Semi-Structured Data

Definition: Data with a partial or incomplete format that can be parsed into structured elements.

Examples:

- XML:

xml

```
<Customer>
  <name>Jane Doe</name>
  <age>30</age>
  <address>123 Main St</address>
</Customer>
<Customer>
  <name>Jane Doe</name>
  <age>30</age>
  <address>123 Main St</address>
```

</Customer>

- JSON:

json

```
{
  "user_id": 12345,
  "name": "John Doe",
  "email": "john@example.com",
  "phone": "+1-555-0101"
}
{
  "user_id": 12345,
  "name": "John Doe",
  "email": "john@example.com",
  "phone": "+1-555-0101"
}
```

- CSV:

csv

```
CustomerID,Name,Age,Address
101,John Doe,30,123 Main St
CustomerID,Name,Age,Address
101,John Doe,30,123 Main St
```

- Log files from web servers or APIs

Why it works: It has a known structure (tags, keys, delimiters), so tools can automatically parse and convert it into structured form.

3. Unstructured Data

Definition: Data that has no predefined format or schema.

Examples:

- Text documents: Emails, chat messages, blog posts, reports
- Images: Photos, screenshots, scanned documents
- Audio/Video: Voice recordings, video files
- Coordinates: GPS data, latitude/longitude
- Social media posts: "Great day out!" with no tags
- PDF documents

Why it fails traditional approaches:

- No fixed schema → hard to query or index
- No relational model → relationships don't exist
- No database engine can parse it directly
- Requires custom parsing or machine learning (OCR, NLP)

Summary Comparison

	Structured	Semi-structured	Unstructured
Format	Fixed schema	Partial format	No format
Parsing	Easy	Easy (with tools)	Difficult
Scalability	Low	Moderate	High
Querying	Indexable	Partially indexable	Needs ML/OCR
Database	Relational	Document stores	No database needed

Real-World Analogy

- Structured: A neatly organized library catalog with fixed book categories.
- Semi-structured: A book in the catalog with some info filled in manually.
- Unstructured: A random book on a shelf with no categories, no title, no author, just "A book about cats."

Why the Distinction Matters

- Structured data → Relational databases, SQL, OLTP systems
- Semi-structured data → Document databases, cloud APIs, GraphQL
- Unstructured data → AI/NLP, image recognition, unstructured analytics

Modern systems (like cloud data warehouses, data lakes, and AI models) are increasingly designed to handle all three types of data together — that's why modern data platforms (Snowflake, BigQuery, AWS S3, Azure Blob Storage) treat all data the same way regardless of its form.

Question 4 (3 Marks)

Explain the role of distributed file systems in big data processing.

Explanation Notes

1	State the main need: very large datasets are split and stored across multiple machines.
2	Mention core benefits: scalability, fault tolerance, parallel access and support for batch processing.
3	Use HDFS or cloud object storage as an example, but do not confuse file storage with relational database storage.
4	Connect storage to processing: data blocks can be processed in parallel by frameworks such as MapReduce or Spark.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

The Role of Distributed File Systems in Big Data Processing

Distributed file systems play a critical role in big data processing by enabling efficient management of massive data volumes across multiple compute nodes.

Their primary roles include:

1. Handling Large Data Volumes – They support storage and retrieval of petabytes or exabytes of data that exceed the capacity of traditional single-node storage.
2. Distributed Processing – Data is stored and processed across many nodes, allowing parallel access and computation rather than sequential processing on a single machine.
3. Scalability and Fault Tolerance – They automatically scale to accommodate growing data loads and ensure data remains accessible even if individual nodes fail, due to replication and redundancy mechanisms.
4. Cost-Effectiveness – By distributing work across multiple nodes, they significantly reduce per-unit storage and compute costs compared to centralized storage solutions.
5. High Availability – They ensure continuous operation and data availability even during partial node failures, which is critical for large-scale workloads.

Question 5 (3 Marks)

List any three major security concerns in big data platforms and briefly explain each.

Explanation Notes

1	Choose three concerns such as unauthorized access, data leakage, privacy violation, weak encryption, poor audit logs or insecure APIs.
2	For each concern, explain the risk and not only the keyword.
3	Add one control where possible: role-based access, encryption, anonymization, masking, logging or monitoring.
4	Show awareness that big data combines many data sources, so security and governance become more complex.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Three Major Security Concerns in Big Data Platforms

Concern	Brief Explanation
1. Data Privacy & Confidentiality	Large datasets can expose sensitive Personally Identifiable Information (PII), such as names, addresses, financial data, or health records. If leaked, this can lead to severe privacy breaches and legal liability.
2. Access Control & Unauthorized Access	Big data systems often handle millions of users and a vast number of queries. Traditional permission controls become impractical; granting or revoking access for millions of users is complex, and insider threats remain a serious risk.
3. Data Integrity & Consistency	In distributed systems, concurrent updates from multiple users can cause data inconsistencies. Without proper concurrency control (e.g., distributed locking, ACID guarantees), duplicate reads, lost updates, and race conditions can occur, corrupting the data.

Question 6 (3 Marks)

What are data appliances and integration tools? Explain their role in big data workflows.

Explanation Notes

1	A data appliance is a specialized hardware/software platform optimized for storage, processing or analytics workloads.
2	Integration tools move, transform or synchronize data between sources, storage systems and analytics layers.
3	Mention ETL/ELT, API connectors, message queues, ingestion tools or workflow schedulers as examples.
4	Explain the workflow: source data -> ingestion -> cleaning/transformation -> storage -> processing -> dashboard/decision.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Data Appliances and Integration Tools in Big Data Workflows

Data Appliances

Data appliances are specialized hardware or software systems designed specifically to handle the processing and storage of large volumes of data. They are designed to manage the high-throughput, high-volume nature of big data processing. Common types include:

- Hadoop clusters (e.g., HDFS, MapReduce)
- Apache HBase (key-value store)
- Apache Cassandra (distributed key-value store)
- In-Memory DBs (e.g., Spark in-memory engines)

Their role is to store and process massive datasets at scale, providing the foundational infrastructure for big data analytics.

Integration Tools

Integration tools are software frameworks or platforms that facilitate the connection, synchronization, and exchange of data between different systems and data sources. They act as the "glue" between disparate systems, enabling seamless data flow. Common examples include:

- Apache Airflow (orchestration of data pipelines)
- Apache NiFi (data flow orchestration)
- Apache Kafka (streaming data integration)
- Apache NiFi / Apache Flume / Logstash (log and stream data transfer)

Their role is to bridge gaps between diverse data sources, ensuring that data from heterogeneous systems (databases, APIs, files, etc.) can be collected, transformed, and fed into the big data processing pipeline.

Question 7 (3 Marks)

Discuss two major challenges organizations face while implementing big data analytics.

Explanation Notes

1	Pick two strong challenges such as data quality, skill gap, infrastructure cost, privacy compliance, integration difficulty or real-time scalability.
2	Explain the cause and impact of each challenge.
3	Suggest a practical mitigation for each challenge, such as governance, cloud scaling, training, access control or phased implementation.
4	Try to connect the challenge to business impact, not only technical difficulty.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Two Major Challenges in Implementing Big Data Analytics

1. Data Silos (Lack of Data Integration)

Large organizations often maintain data silos — data residing in isolated systems (e.g., CRM, ERP, legacy databases) without seamless connectivity. To implement big data analytics, organizations must:

- Identify and connect multiple data sources,
- Emerge a unified data warehouse,
- Transform data into a single, accessible format.

Without addressing this first, the entire analytics stack cannot be effectively deployed.

2. Data Quality and Accuracy

Big data analytics rely on accurate data to produce meaningful insights. However, organizations frequently face data quality issues, such as:

- Incomplete, outdated, or inconsistent data,
- Inaccurate measurements,
- Missing or erroneous records.

Since big data analytics depend heavily on data-driven decisions, poor data quality directly undermines the reliability and credibility of insights derived from the data.

Summary: The two primary challenges are (1) data silos, which prevent data from being collected and shared across systems, and (2) data quality, which ensures the insights derived from big data are trustworthy. Both must be addressed before big data analytics can deliver meaningful business value.

Question 8 (3 Marks)

Write short notes on any one domain application of big data analytics.

Explanation Notes

1	Select one domain: healthcare, retail, agriculture, finance, education, smart city, transport or cybersecurity.
2	Identify data sources in that domain.

3	Explain how analytics converts raw data into decisions: prediction, recommendation, alert, optimization or risk detection.
4	Mention one expected outcome such as reduced waiting time, fraud detection, crop advisory or improved user experience.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Domain Application of Big Data Analytics

Healthcare: Predictive Disease Detection

Big data analytics is revolutionizing healthcare by enabling predictive disease detection. By analyzing vast datasets from electronic health records, genomic data, lifestyle factors, and medical imaging, organizations can identify patterns that predict the onset of diseases like cancer or cardiovascular conditions before they become apparent.

This is critical because traditional diagnostics often occur too late in the disease progression. By detecting patterns early—such as abnormal biomarkers in tumor markers or unusual activity patterns in patient records—healthcare providers can implement targeted interventions, reduce mortality rates, and significantly improve patient outcomes. This leads to better patient management, reduced treatment costs, and more efficient healthcare system operations.

Question 9 (3 Marks)

How do business drivers influence the adoption of big data solutions?

Explanation Notes

1	Business drivers are reasons that push an organization to adopt big data, such as cost reduction, personalization, speed, risk control or better decisions.
2	Explain how each selected driver creates a technical requirement.
3	Example: personalization requires behavior tracking and recommendation analytics; risk control requires fraud detection and monitoring.
4	Conclude that big data adoption must be justified by measurable business value.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Business Drivers for Big Data Adoption

1. Improving Customer Experience and Engagement Businesses adopt big data analytics to gain deeper insights into customer behavior, preferences, and preferences. By analyzing customer interactions, purchasing patterns, and engagement metrics, organizations can personalize experiences, drive sales, and increase customer loyalty. This directly impacts revenue and retention.
2. Enhancing Decision-Making and Strategic Planning Data-driven decision-making becomes more reliable when backed by large datasets. Big data analytics enables leaders to make informed strategic decisions by analyzing historical and real-time data, which is essential for long-term planning and resource allocation.
3. Gaining a Competitive Advantage In today's fast-evolving market, businesses adopt big data to stay ahead of competitors. By identifying market trends early, uncovering new opportunities, and innovating quickly, organizations can differentiate themselves and capture market share.

Question 10 (3 Marks)

Explain the importance of big data analytics in decision making.

Explanation Notes

1	Explain that analytics helps convert raw, high-volume data into useful patterns, predictions, alerts or recommendations.
2	Differentiate descriptive, diagnostic, predictive and prescriptive decision support if space permits.
3	Use one example: dashboard for hospital beds, forecast for demand, fraud alert for bank, traffic prediction for smart city.
4	End with the idea that decisions become faster, evidence-based and scalable.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Importance of Big Data Analytics in Decision Making

1. **Informed Decision-Making** Big data analytics enables organizations to make decisions based on actual data rather than intuition or guesswork. By analyzing historical and real-time data, leaders can choose the most effective path forward, reducing uncertainty and risk in strategic choices.
2. **Competitive Advantage** Organizations that harness big data analytics can stay ahead of competitors. By identifying emerging trends, market gaps, and customer preferences before others, businesses can innovate faster, capture market share, and respond to competition with precision.
3. **Improved Efficiency and Cost Reduction** Data analytics helps identify bottlenecks, optimize processes, and eliminate waste. By understanding where resources are being wasted and where improvements can be made, organizations can reduce operational costs and allocate resources more effectively.
4. **Personalized Customer Engagement** By understanding what drives customer behavior, businesses can tailor products, services, and communications to individual customers. This leads to higher satisfaction, increased loyalty, and improved revenue.

In short, big data analytics transforms decision-making from subjective to objective, enabling organizations to navigate uncertainty and capitalize on opportunities more effectively.

Submission Checklist

Checklist Item	Status	Remarks
Student details completed.	Done / Pending	done
All 10 questions answered.	Done / Pending	done
Examples are connected to cloud/capstone/domain context.	Done / Pending	done
Security/privacy points are included wherever relevant.	Done / Pending	done
Document exported as PDF if requested.	Done / Pending	done
GitHub link shared through Google Classroom.	Done / Pending	done